Physics-Prior Latent World Models for Reliable Long-Horizon UAV Prediction under Out-of-Distribution Wind Disturbances

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***Abstract*—Learned world models provide a mechanism for predicting future states without repeatedly interacting with the physical environment. However, recursive prediction can accumulate error over long horizons, and this problem can become more severe when the operating conditions differ from those observed during training. Quadrotor UAVs provide a particularly interesting setting because their nominal rigid-body dynamics are known, while aerodynamic effects, wind disturbances, actuator mismatch, and other unmodeled effects remain difficult to describe analytically. This thesis investigates whether incorporating known quadrotor dynamics as a physical prior into a latent world model can improve the reliability of long-horizon prediction under out-of-distribution (OOD) wind disturbances. The proposed model decomposes the continuous-time dynamics into a known physics component and a learned residual,**

$$\dot{s}_t = f_{\text{phys}}(s_t, u_t) + d_\theta(h_t, s_t, u_t),$$

where s_t denotes the physical UAV state and h_t denotes a learned latent memory state. The resulting dynamics are integrated numerically and fed back into the latent transition to enable open-loop imagination without future observations.

The primary evaluation compares a purely learned latent dynamics model with the proposed physics-prior residual model under identical training data and evaluation conditions. The main evaluation criterion is not only one-step prediction accuracy but the rate of error growth during recursive rollouts. A reliable imagination horizon is defined as the maximum prediction horizon for which the mean position prediction error remains below a specified tolerance. Experiments will use simulated quadrotor trajectories with controlled wind distributions, with wind magnitude extrapolation serving as the primary OOD setting and directional and transient disturbances serving as secondary stress tests. The study will also investigate when a physical prior helps and when model mismatch can reduce its benefit. Visual observations, uncertainty estimation, and model-predictive control are considered extensions rather than requirements of the core thesis.

***Index Terms*—UAV, quadrotor, world models, latent dynamics, physics prior, residual learning, long-horizon prediction, distribution shift, wind disturbances, reliable imagination.**

I. Introduction and Motivation

A. Background

Unmanned Aerial Vehicles (UAVs), particularly quadrotors, operate according to well-established nonlinear rigid-body dynamics. A simplified rigid-body model can be represented by the translational and rotational equations

$$\dot{\mathbf{p}} = \mathbf{v}, \tag{1}$$

$$m\dot{\mathbf{v}} = m\mathbf{g} + \mathbf{R}\mathbf{f}_T + \mathbf{f}_{\text{ext}}, \tag{2}$$

$$\dot{\mathbf{R}} = \mathbf{R}[\boldsymbol{\omega}]_\times, \tag{3}$$

$$\mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) + \boldsymbol{\tau}_{\text{ext}}, \tag{4}$$

where $\mathbf{p}$ and $\mathbf{v}$ denote position and velocity, $\mathbf{R} \in SO(3)$ denotes the attitude rotation matrix, $\boldsymbol{\omega}$ denotes body-frame angular velocity, m is the vehicle mass, and $\mathbf{J}$ is the body-frame inertia matrix. The vectors $\mathbf{f}_T$ and $\boldsymbol{\tau}$ denote the total thrust and control torque, respectively. For an idealized quadrotor, the total thrust magnitude can be related to the rotor angular velocities by

$$f_T = \sum_{i=1}^{4} k_f \Omega_i^2, \tag{5}$$

where k_f is the thrust coefficient and Ω_i is the angular velocity of rotor i. The corresponding body torques are determined by the quadrotor geometry, rotor configuration, and motor rotation directions and can be represented through a control-allocation matrix. These equations provide substantial prior knowledge about how the vehicle evolves. However, the nominal rigid-body model does not perfectly describe all effects encountered by a real or high-fidelity simulated UAV. Important sources of mismatch include

- aerodynamic drag,
- wind and turbulence,
- actuator dynamics and delays,
- uncertainty in mass and inertia,
- aerodynamic coupling,
- rotor and ground effects,
- modeling and parameter uncertainty.

Consequently, the actual system dynamics can be viewed as the combination of the nominal physics model and the effects that are not captured by the nominal model. This decomposition motivates hybrid physics–learning approaches in which known dynamics are handled analytically while a neural model learns the remaining discrepancy. Importantly, the learned residual in this thesis is not interpreted as a direct estimate of wind velocity. Instead, it represents the net effect

of wind, aerodynamic mismatch, actuator mismatch, parameter uncertainty, and other effects not captured by the nominal model.

B. Why Learned World Models?

A world model learns an internal representation of an environment and a transition model that can be recursively used to predict future states. A generic deterministic latent transition can be written as

$$z_{t+1} = F_\theta(z_t, u_t). \tag{6}$$

A stochastic formulation may instead represent

$$p(z_{t+1} \mid z_t, u_t). \tag{7}$$

The main advantage of such models is that future trajectories can be generated internally without physically executing every candidate action. This is particularly attractive for UAVs because

- physical experiments can be expensive,
- dangerous conditions are difficult to collect,
- long-horizon decision making requires future prediction,
- imagined trajectories can be generated much faster than physical experiments,
- latent state prediction avoids the requirement to generate photorealistic future images.

However, recursive prediction introduces an important problem. If

$$\hat{s}_{t+1} = F_\theta(\hat{s}_t, u_t), \tag{8}$$

then the prediction at time $t+1$ becomes the input for predicting $t+2$. Therefore,

$$\hat{s}_{t+1} \to \hat{s}_{t+2} \to \cdots \to \hat{s}_{t+H}. \tag{9}$$

where H is the horizon length. Even a small one-step error can consequently affect all subsequent predictions. A model that performs well at one-step prediction may therefore become unreliable at long imagination horizons.

C. Distribution Shift

The problem becomes more challenging when the conditions during evaluation differ from those used during training. Let the wind disturbance during training be sampled from

$$\mathbf{w} \sim P_{\text{train}}(\mathbf{w}). \tag{10}$$

During evaluation, the model may encounter

$$\mathbf{w} \sim P_{\text{test}}(\mathbf{w}), \qquad P_{\text{test}} \neq P_{\text{train}}. \tag{11}$$

For UAVs, distribution shifts can arise from

- stronger wind magnitudes,
- previously unseen wind directions,
- different temporal characteristics,
- sudden gusts,
- moderate changes in vehicle parameters.

A purely learned model must extrapolate the complete dynamics under these conditions. In contrast, some structural relationships, such as gravity, thrust generation, translational kinematics, and rotational dynamics, remain valid even when the disturbance distribution changes.

This motivates the central hypothesis of this thesis: if the known physics explains a sufficiently large portion of the vehicle dynamics, then the learned model only needs to represent the remaining discrepancy. This may reduce recursive error growth under distribution shift.

D. Research Problem

The central problem addressed by this thesis is:

> *Can known quadrotor physical dynamics, incorporated as a prior in a latent world model, improve the reliability of recursive long-horizon prediction under out-of-distribution wind disturbances?*

The thesis deliberately focuses on the predictive dynamics problem rather than requiring successful autonomous navigation or reinforcement learning.

The central scientific question can therefore be stated more concisely as

> *Does incorporating known UAV physics slow the growth of prediction error during long-horizon imagination under OOD wind?*

II. Literature Review

A. Physics-Based Quadrotor Dynamics

Quadrotor dynamics have traditionally been modeled using first-principles rigid-body mechanics. Such models provide physically interpretable predictions and are widely used in simulation and control. Their main limitation is that the analytical model is only an approximation of the complete vehicle dynamics. Aerodynamic forces, wind, actuator dynamics, and uncertain parameters can introduce discrepancies between the model and the actual system. This motivates augmenting analytical models with data-driven components rather than attempting to replace the physical model entirely.

B. Physics-Informed Learning for UAV Dynamics

Physics-informed learning incorporates known physical structure into neural models. Gu et al. [5] investigated physics-informed neural modeling for quadrotor dynamics and incorporated physical information into the learning process. Their results under aerodynamic effects and periodic wind demonstrate that physical knowledge can improve learned UAV dynamics and physical consistency. Such approaches demonstrate the value of physical priors, but a physics-informed loss and a physics-based transition are not necessarily equivalent. In a loss-based formulation, physical knowledge may constrain the solution during training while the learned transition itself remains a neural approximation. The present thesis instead investigates the more explicit formulation in which the known physical dynamics directly participate in the transition used during imagination.

C. Hybrid Physics–Learning and Residual Dynamics

A closely related research direction combines a first-principles model with a learned residual. The general formulation is

$$\dot{s}_t = f_{\text{phys}}(s_t, u_t) + d_\theta(s_t, u_t). \tag{12}$$

The physical model explains the known portion of the dynamics, while the neural component learns the discrepancy. NeuroBEM by Bauersfeld et al. [6] provides a representative example of this approach for quadrotor modeling. The method combines a first-principles aerodynamic model with learned components to capture effects that are difficult to model analytically. The results demonstrate that hybrid modeling can provide advantages over both purely analytical and purely learned alternatives. More recent work has also applied physics-guided residual learning to UAV trajectory prediction. Islam et al. [8] combine physics-based trajectory prediction with learned residual corrections and physical feasibility constraints. These studies establish that residual learning is a viable mechanism for combining analytical dynamics with learned corrections. However, they do not by themselves answer the question of how such hybrid dynamics affect the reliability of a recurrent latent world model during long open-loop imagination.

D. UAV Dynamics under Wind and Distribution Shift

Wind represents an important source of model uncertainty in UAV dynamics. It can change both the magnitude and direction of forces acting on the vehicle and can produce significant differences between training and deployment conditions. PI-WAN by Wang et al. [7] investigates physics-informed wind-adaptive quadrotor dynamics prediction under varying wind, payload, and external perturbation conditions. The work demonstrates that incorporating physical information can improve prediction and closed-loop tracking under previously unseen conditions. This work is particularly relevant because it establishes the value of physics-informed modeling under environmental distribution shift. However, its primary objective is wind-adaptive dynamics prediction and control rather than measuring the reliable imagination horizon of a latent world model.

E. Latent World Models

World Models [1] established the idea of learning compact internal representations of environments together with predictive dynamics. PlaNet [2] introduced a recurrent stochastic latent state-space model for planning, while Dreamer [3] demonstrated that behavior can be learned using trajectories imagined inside a learned latent model. DreamerV3 further demonstrated the scalability of the world-model paradigm across diverse environments [4]. A generic latent world model can be represented as

$$e_t = E_\theta(o_t), \tag{13}$$

followed by

$$h_{t+1} = F_\theta(h_t, e_t, u_t). \tag{14}$$

The model can subsequently be rolled forward without future observations. The central limitation relevant to this thesis is recursive error accumulation. The quality of a long imagined trajectory depends not only on the one-step transition but also on the stability of repeated application of the learned transition.

F. World Models for UAVs

Recent work has increasingly explored learned world models for aerial navigation and UAV prediction. Navigation-oriented models such as AirDreamer [10] and WorldFly [11] world-model approaches demonstrate the potential of learned predictive representations for aerial decision making. More recent work such as SkyJEPA [9] specifically investigates long-horizon world modeling for quadrotors. This is important because it demonstrates that long-horizon learned predictive models for UAVs are themselves an active research direction. The present thesis therefore does not claim that UAV world modeling is new. Instead, it studies a narrower question: whether explicit quadrotor physical structure can improve the reliability of a latent world model when it is recursively rolled forward under OOD disturbances.

G. Long-Horizon Reliability

A key limitation of evaluating learned dynamics using only one-step metrics is that one-step accuracy does not necessarily predict long-horizon reliability [12].

For recursive prediction,

$$\hat{s}_{t+k} = F_\theta^{(k)}(s_t, u_{t:t+k-1}), \tag{15}$$

and the prediction error may increase with k. Therefore, a meaningful evaluation should examine prediction error as a function of horizon,

$$E(H) = \frac{1}{H}\sum_{k=1}^{H} e_{t+k}, \tag{16}$$

rather than reporting only one-step RMSE. For UAVs, position, velocity, and attitude should be evaluated separately because they have different physical units and scales.

H. Research Gap

The existing literature establishes several important facts:

1) known physical structure can improve learned UAV dynamics;
2) residual learning can capture discrepancies between analytical and observed dynamics;
3) physics-informed approaches can improve robustness under wind and environmental changes;
4) latent world models can support recursive imagined rollouts;
5) UAV world models can support long-horizon predictive reasoning.

However, these directions have largely been evaluated separately. In particular, there is comparatively limited systematic analysis of the following controlled comparison:

- **Pure learned latent dynamics**
- **Physics-prior latent dynamics**

when both are evaluated using recursive long-horizon prediction under controlled OOD wind disturbances. The contribution of this thesis is therefore not the claim that physics-informed learning, residual dynamics, or latent world models are individually novel. Instead, the thesis aims to provide a controlled investigation of their interaction and determine whether the physical prior changes the rate of long-horizon error growth under distribution shift.

III. Research Questions and Hypotheses

The thesis is organized around one primary research question and a small number of secondary questions.

A. Primary Research Question

RQ1: Does incorporating known quadrotor dynamics into a latent world model improve long-horizon predictive reliability under OOD wind disturbances?

B. Secondary Research Questions

RQ2: Under which OOD wind conditions does the physics prior provide the largest benefit?

RQ3: How sensitive is the benefit of the physics prior to the accuracy of the nominal physical model and the amount of training data?

RQ4: Can predictive uncertainty, if incorporated as an extension, provide a useful indicator of the horizon beyond which imagined trajectories become unreliable?

Vision is not a primary research question. It is treated as an optional extension after the core proprioceptive model has been established.

C. Primary Hypothesis

The primary hypothesis is

$$\boxed{H_1: \quad E_p^{\text{phys}}(H) < E_p^{\text{pure}}(H)} \tag{17}$$

The physics-prior model is expected to exhibit slower multi-step prediction error growth under OOD wind disturbances.

An important secondary hypothesis is conditional:

$$\boxed{H_2: \quad E_p^{\text{phys}}(H;\delta) \geq E_p^{\text{pure}}(H;\delta)} \tag{18}$$

for sufficiently large δ, where δ denotes the magnitude of nominal physics-model mismatch. The benefit of the physics prior is therefore expected to decrease as the nominal physics model becomes sufficiently inaccurate. The thesis does not assume that H_1 must hold in every condition. A negative or conditional result is scientifically meaningful if the experiments identify the conditions under which the physical prior fails.

IV. Progress Achieved: Proposed Method

A. Physical State

The physical state of the UAV is defined as

$$s_t = [\mathbf{p}_t,\, \mathbf{v}_t,\, \mathbf{R}_t,\, \boldsymbol{\omega}_t]\,, \tag{19}$$

where

$$\mathbf{p}_t \in \mathbb{R}^3, \qquad \mathbf{v}_t \in \mathbb{R}^3, \tag{20}$$

$$\mathbf{R}_t \in \mathrm{SO}(3), \qquad \boldsymbol{\omega}_t \in \mathbb{R}^3. \tag{21}$$

Quaternion representation may be used for storage and learning, while the physical dynamics and attitude error are represented using rotation matrices or an equivalent manifold-aware representation. The control input is

$$u_t = [\Omega_{1,t}, \Omega_{2,t}, \Omega_{3,t}, \Omega_{4,t}], \tag{22}$$

or the equivalent motor/thrust command provided by the simulator.

B. Latent State

The latent state h_t represents information needed to predict the future that is not fully captured by the instantaneous physical state. It is updated from sensor observations through

$$h_t = F_\theta\left(h_{t-1}, e_t, u_{t-1}\right), \tag{23}$$

where e_t is an encoded observation. For the core thesis, the observation is restricted to proprioceptive information:

$$e_t = E_\theta\left(a_t^{\text{IMU}}, \boldsymbol{\omega}_t^{\text{IMU}}, u_t\right). \tag{24}$$

This keeps the main problem focused on dynamics learning rather than visual representation learning.

C. Physical State Decoder

The latent state is mapped to a physical state estimate through

$$\hat{s}_t = D_\theta(h_t). \tag{25}$$

During training, the decoder is supervised using simulator state measurements. The predicted state is therefore explicitly interpretable in terms of position, velocity, attitude, and angular velocity.

D. Physics-Prior Transition

The central model decomposes the continuous-time dynamics into known physics and learned residual dynamics:

$$\boxed{\dot{\hat{s}}_t = f_{\text{phys}}(\hat{s}_t, u_t) + d_\theta(h_t, \hat{s}_t, u_t).} \tag{26}$$

The first term represents the known nominal quadrotor dynamics. The second term represents effects that are not adequately captured by the nominal model. The learned residual captures effects not represented by the nominal physics model:

$$d_\theta \approx d_{\text{wind}} + d_{\text{aero}} + d_{\text{param}} + d_{\text{other}}. \tag{27}$$

Here, d_{wind}, d_{aero}, d_{param}, and d_{other} denote the contributions of wind effects, aerodynamic mismatch, parameter mismatch, and other unmodeled effects, respectively. This avoids requiring the network to identify wind velocity from available observations.

E. Numerical Integration

The continuous-time dynamics are converted into a discrete transition using a numerical integrator, preferably fourth-order Runge–Kutta:

$$\tilde{s}_{t+1} = \text{RK4}\left(\hat{s}_t, f_{\text{phys}}(\hat{s}_t, u_t) + d_\theta(h_t, \hat{s}_t, u_t), \Delta t\right). \tag{28}$$

The numerical integration step provides a structured transition rather than asking the neural network to learn the complete state evolution directly.

F. Latent–Physical Coupling

After predicting the next physical state, the predicted state is fed back into the latent dynamics:

$$\boxed{\hat{h}_{t+1} = F_\theta\left(h_t, \phi(\tilde{s}_{t+1}), u_t\right),} \tag{29}$$

where $\phi(\cdot)$ denotes a state embedding.

The overall architecture is therefore

$$\boxed{h_t \rightarrow \text{physical transition} \rightarrow \tilde{s}_{t+1} \rightarrow h_{t+1}.} \tag{30}$$

The latent state provides memory, while the physical transition provides the structured evolution of the UAV.

G. Open-Loop Imagination

During observation, the latent state can use sensor measurements:

$$h_t = F_\theta(h_{t-1}, e_t, u_{t-1}). \tag{31}$$

During imagination, future observations are unavailable. The model must therefore recursively use its own predictions:

$$h_t \rightarrow \tilde{s}_{t+1} \rightarrow h_{t+1} \rightarrow \tilde{s}_{t+2} \rightarrow h_{t+2} \rightarrow \cdots \tag{32}$$

$$\rightarrow \tilde{s}_{t+H}. \tag{33}$$

This open-loop rollout is the primary object evaluated by the thesis.

V. Future Plans: Training Methodology

A. Dataset Generation

The initial dataset will be generated in a physics-based simulator such as Isaac Sim / Isaac Lab / OmniVerse.

Each trajectory will contain, where available,

- position,
- linear velocity,
- attitude,
- angular velocity,
- IMU acceleration,
- IMU angular velocity,
- motor commands or rotor speeds,
- wind disturbance,
- external disturbance force,
- simulation parameters.

Wind measurements and disturbance forces are recorded for analysis and for oracle experiments but are not required as inputs to the core model.

B. Trajectory Generation

The dataset will consist of trajectories generated under diverse control sequences and wind conditions.

The control trajectories should include

- hovering,
- translational motion,
- acceleration and deceleration,
- yaw motion,
- moderate attitude changes,
- combined translational and rotational motion.

The dataset should be split by complete trajectories rather than by individual frames in order to avoid temporal leakage.

C. Wind Distribution

The primary OOD experiment will use wind-magnitude extrapolation.

For example,

$$|\mathbf{w}|_{\text{train}} \in [0, 4]\ \text{m/s}, \tag{34}$$

while

$$|\mathbf{w}|_{\text{test}} \in (4, 8]\ \text{m/s}. \tag{35}$$

The exact ranges will be selected after verifying simulator stability and the operating envelope of the chosen UAV.

Secondary OOD conditions include

1) unseen wind directions,
2) different temporal characteristics,
3) sudden gust disturbances.

Physics-parameter mismatch is treated primarily as an ablation rather than as a fourth independent main OOD axis.

D. Physics Self-Consistency Study

Before training the neural models, the nominal physics model will be evaluated against simulator ground truth.

Case	Condition	Purpose
A	No wind	Baseline physics accuracy
B	Training-range wind	Disturbance-induced error
C	Parameter perturbation	Physics-model mismatch

This experiment is important because the usefulness of a physics prior depends on the quality of the physical model.

E. Training Stages

The model will be trained progressively.

a) Stage 1: Physics-only model.: Evaluate

$$\dot{s} = f_{\text{phys}}(s, u) \tag{36}$$

without learning a residual.

b) Stage 2: Pure neural baseline.: Train a neural transition model

$$\hat{s}_{t+1} = F_\theta^{\text{pure}}(\hat{s}_t, u_t, h_t). \tag{37}$$

The model should have approximately comparable parameter capacity to the proposed residual model.

c) Stage 3: Physics + residual.: Train

$$\dot{\hat{s}} = f_{\text{phys}} + d_\theta. \tag{38}$$

d) Stage 4: Physics + residual without latent memory.: Use

$$d_\theta(s_t, u_t) \tag{39}$$

instead of

$$d_\theta(h_t, s_t, u_t). \tag{40}$$

This ablation determines whether recurrent latent memory provides additional benefit.

e) Stage 5: Long-horizon training.: Gradually increase the rollout horizon during training rather than immediately optimizing extremely long trajectories.

A possible curriculum is

$$H_{\text{train}} = 1 \to 5 \to 10 \to 20 \to 50. \tag{41}$$

The final horizon will be selected based on computational cost and initial experimental stability.

F. Prediction Loss

The physical-state prediction loss is decomposed by state component:

$$\mathcal{L}_{\text{state}} = \lambda_p \mathcal{L}_p + \lambda_v \mathcal{L}_v + \lambda_R \mathcal{L}_R + \lambda_\omega \mathcal{L}_\omega. \tag{42}$$

Position loss:

$$\mathcal{L}_p = \|\hat{\mathbf{p}}_t - \mathbf{p}_t\|_2^2. \tag{43}$$

Velocity loss:

$$\mathcal{L}_v = \|\hat{\mathbf{v}}_t - \mathbf{v}_t\|_2^2. \tag{44}$$

Attitude loss:

$$\mathcal{L}_R = d_{\text{SO(3)}}(\hat{\mathbf{R}}_t, \mathbf{R}_t)^2. \tag{45}$$

Angular-velocity loss:

$$\mathcal{L}_\omega = \|\hat{\boldsymbol{\omega}}_t - \boldsymbol{\omega}_t\|_2^2. \tag{46}$$

G. Multi-Step Training Loss

For an imagination horizon H, the multi-step loss is

$$\mathcal{L}_{\text{roll}} = \sum_{k=1}^{H} \gamma^{k-1} \mathcal{L}_{\text{state}}(\hat{s}_{t+k}, s_{t+k}), \tag{47}$$

where γ controls the relative importance of later predictions. The final training objective is

$$\mathcal{L} = \mathcal{L}_{\text{state}} + \lambda_{\text{roll}} \mathcal{L}_{\text{roll}}. \tag{48}$$

A residual regularization term may be investigated as an ablation:

$$\mathcal{L}_{\text{smooth}} = \sum_t \|d_{t+1} - d_t\|_2^2. \tag{49}$$

This term will not be assumed to be universally appropriate because rapid disturbances can legitimately produce rapidly changing residuals.

VI. Experimental Baselines

The main experiment is designed to isolate the contribution of the physical prior.

Model	Description
B1	Physics-only model
B2	Pure neural latent dynamics
B3	Physics + learned residual + latent memory
B4	Physics + residual without latent memory
B5	Residual-only model
B6	Physics + oracle disturbance

The central comparison is

$$\boxed{\text{B2} \quad \text{vs.} \quad \text{B3}.} \tag{50}$$

Both models should use the same training trajectories, observation history, prediction targets, optimizer protocol, and approximately matched model capacity.

The additional baselines serve different purposes:

- B1 measures the performance of the analytical model alone.
- B4 isolates the contribution of recurrent latent memory.
- B5 determines whether explicitly separating physics from the learned component is beneficial.
- B6 provides an approximate upper-bound/reference condition when the disturbance is available.

VII. OOD Evaluation Protocol

A. Primary OOD Regime: Wind Magnitude

The primary experiment evaluates extrapolation beyond the training wind range:

$$P_{\text{train}} : |\mathbf{w}| \in [0, W_{\text{train}}], \tag{51}$$

while

$$P_{\text{test}} : |\mathbf{w}| \in (W_{\text{train}}, W_{\text{test}}]. \tag{52}$$

The model is not retrained on the OOD test distribution.

B. Secondary OOD Regime: Direction

The second experiment evaluates wind directions that are not present, or are underrepresented, during training. This tests whether the physical structure generalizes to a change in the direction of the disturbance.

C. Secondary OOD Regime: Temporal Disturbances

A further stress test changes the temporal structure of the disturbance, such as turbulence frequency or correlation time. The purpose is not to establish universal turbulence robustness but to determine whether the observed advantage of the physical prior persists under a qualitatively different disturbance process.

D. Secondary OOD Regime: Gusts

Sudden transient wind disturbances are introduced as another stress test. The experiment evaluates whether a model trained primarily on continuous disturbances can maintain reliable predictions when exposed to short-duration disturbances.

E. Physics-Model Mismatch

The nominal model parameters are perturbed during evaluation:

$$m' = (1 + \delta_m)m, \tag{53}$$

$$\mathbf{J}' = \mathbf{J} + \Delta\mathbf{J}. \tag{54}$$

Aerodynamic parameters may also be perturbed. This experiment tests an important limitation: A physics prior is useful only if its structural assumptions remain sufficiently accurate.

VIII. Evaluation Metrics

A. One-Step Prediction

One-step prediction is evaluated using separate physical metrics.

Position:

$$\text{RMSE}_p = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{p}}_i - \mathbf{p}_i\|_2^2}. \tag{55}$$

Velocity:

$$\text{RMSE}_v = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{v}}_i - \mathbf{v}_i\|_2^2}. \tag{56}$$

Attitude:

$$e_R = d_{\text{SO}(3)}(\hat{\mathbf{R}}, \mathbf{R}). \tag{57}$$

Angular velocity:

$$\text{RMSE}_\omega = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|\hat{\boldsymbol{\omega}}_i - \boldsymbol{\omega}_i\|_2^2}. \tag{58}$$

B. Long-Horizon Position Error

Because the primary concern is whether the predicted trajectory remains physically useful over long horizons, position is used as the primary reliability quantity.

For a trajectory starting at time t, define

$$\boxed{E_p(H) = \frac{1}{H}\sum_{k=1}^{H} \left\|\hat{\mathbf{p}}_{t+k} - \mathbf{p}_{t+k}\right\|_2.} \tag{59}$$

Velocity and attitude errors are reported separately:

$$E_v(H) = \frac{1}{H}\sum_{k=1}^{H} \left\|\hat{\mathbf{v}}_{t+k} - \mathbf{v}_{t+k}\right\|_2, \tag{60}$$

and

$$E_R(H) = \frac{1}{H}\sum_{k=1}^{H} d_{\text{SO}(3)}(\hat{\mathbf{R}}_{t+k}, \mathbf{R}_{t+k}). \tag{61}$$

This avoids combining quantities with incompatible units into a single state error.

C. Reliable Imagination Horizon

For a chosen position-error tolerance ϵ, define

$$H_\epsilon = \max\{H : E_p(k) \le \epsilon,\ \forall k \le H\}. \tag{62}$$

This gives a direct interpretation: H_ϵ is the maximum horizon for which the predicted UAV position remains within the specified average error tolerance. Rather than reporting only one threshold, the experiment will evaluate multiple values of ϵ. For example,

$$\epsilon \in \{0.25, 0.5, 1.0\}\ \text{m}, \tag{63}$$

subject to the scale and operating range of the selected UAV.

D. Reliability Curve

A second metric measures the fraction of trajectories that remain reliable at each horizon.

For trajectory i,

$$E_p^{(i)}(H) = \frac{1}{H}\sum_{k=1}^{H}\left\|\hat{\mathbf{p}}_{t+k}^{(i)} - \mathbf{p}_{t+k}^{(i)}\right\|_2. \tag{64}$$

Define

$$\boxed{R_\epsilon(H) = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}\left[E_p^{(i)}(H) \le \epsilon\right].} \tag{65}$$

Thus $R_\epsilon(H)$ represents the fraction of trajectories whose average position error remains below ϵ at horizon H. This provides a more robust characterization than a single average error value.

IX. Ablation Studies

The following ablations will be performed where computationally feasible.

A. Physics Ablation

Compare

$$\text{Pure neural} \quad \text{vs.} \quad \text{Physics + residual}. \tag{66}$$

This is the primary ablation.

B. Latent-Memory Ablation

Compare

$$d_\theta(s_t, u_t) \tag{67}$$

against

$$d_\theta(h_t, s_t, u_t). \tag{68}$$

This determines whether temporal memory is useful for representing history-dependent disturbances and model discrepancies.

C. Training-Data Ablation

Train the models using different fractions of the available training data:

$$\{25\%, 50\%, 100\%\}. \tag{69}$$

This tests whether the physical prior provides a sample-efficiency advantage.

D. Physics-Mismatch Ablation

The nominal physical parameters are gradually perturbed. The resulting relationship

$$\text{physics mismatch} \rightarrow \text{prediction performance} \tag{70}$$

will be analyzed to determine when the physical prior becomes harmful.

X. Optional Vision Extension

The core thesis does not require visual observations. After the proprioceptive model is established, RGB or depth observations may be added through a visual encoder:

$$z_t^V = E_V(I_t). \tag{71}$$

The proprioceptive representation is

$$z_t^P = E_P(a_t^{\mathrm{IMU}}, \boldsymbol{\omega}_t^{\mathrm{IMU}}, u_t). \tag{72}$$

The two representations may be fused:

$$e_t = \mathrm{Fuse}(z_t^V, z_t^P). \tag{73}$$

The purpose of this extension is only to determine whether visual information provides predictive information beyond proprioception. It is not necessary for demonstrating the central thesis hypothesis.

XI. Experimental Analysis

The central experimental analysis will compare prediction-error growth as the imagination horizon increases. For each model, plots will be generated for

$$E_p(H) \quad \text{versus} \quad H. \tag{74}$$

The primary comparison will be made under

1) in-distribution wind,
2) mild OOD wind,
3) strong OOD wind,
4) unseen wind direction,
5) transient gust conditions.

The expected result, if the primary hypothesis is supported, is

$$E_p^{\mathrm{physics+residual}}(H) < E_p^{\mathrm{pure}}(H) \tag{75}$$

for sufficiently large H under OOD conditions.

Equivalently,

$$H_\epsilon^{\mathrm{physics+residual}} > H_\epsilon^{\mathrm{pure}}. \tag{76}$$

However, the experiments will not assume that this inequality holds universally. The benefit of the physics prior is expected to depend on the magnitude of the wind distribution shift, the amount and diversity of training data, and the degree of mismatch between the nominal physics model and the true system dynamics.

XII. Expected Scientific Outcomes

Three broad outcomes are possible.

A. Physics Provides a Consistent Benefit

If

$$H_\epsilon^{\mathrm{physics+residual}} > H_\epsilon^{\mathrm{pure}} \tag{77}$$

across multiple OOD conditions, the results would provide evidence that incorporating known UAV physics improves long-horizon predictive reliability.

B. Physics Provides a Conditional Benefit

The physical prior may help under moderate distribution shifts or limited data but provide little advantage under severe mismatch. This would indicate that physical priors are beneficial only within an appropriate operating regime.

C. Physics Does Not Provide a Benefit

The physics-prior model may perform similarly to or worse than the purely learned model.

This could occur if

- the nominal physics model is inaccurate,
- the learned residual is difficult to optimize,
- the distribution shift is too severe,
- the latent representation is inadequate,
- the purely learned model has sufficient data to learn the required dynamics.

Such a result would still be scientifically useful because it would identify limitations of physics priors in learned UAV world models.

XIII. Scope and Limitations

A. Required Components

The minimum viable thesis consists of

- UAV simulation,
- nominal quadrotor rigid-body dynamics,
- proprioceptive observations,
- learned residual dynamics,
- recurrent latent state,
- numerical physical integration,
- open-loop imagination,
- controlled OOD wind experiments,
- long-horizon prediction evaluation,
- reliable imagination horizon analysis.

B. Extensions

The following are extensions if sufficient time remains:

- uncertainty estimation,
- visual observations,
- multimodal sensor fusion,
- additional vehicle-parameter shifts,
- uncertainty-aware model predictive control.

C. Not Required

The following are explicitly outside the core thesis:

- real-world flight testing,
- perfect wind estimation,
- wind-state identification,
- photorealistic future-frame generation,
- autonomous navigation,
- reinforcement-learning policy training,
- full autonomous MPC system,
- generalization across arbitrary UAV morphologies.

The thesis may use multiple configurations of the same quadrotor family as an additional robustness experiment, but broad morphology generalization is not a primary claim.

XIV. Feasibility

The proposed work is designed to be achievable within an approximately eight-month M.Tech thesis timeline.

The primary reasons for feasibility are:

1) The dataset can be generated entirely in simulation or use existing datasets.
2) The required physical state is directly available from the simulator.
3) The core model does not require large-scale image generation.
4) The physical model reduces the amount of dynamics that the neural network must learn.
5) The main experiments use a small number of controlled OOD regimes.
6) The core thesis does not depend on successful RL or autonomous navigation.

The core result should be achievable before the final two months, leaving time for ablations, uncertainty experiments, analysis, and thesis writing.

XV. Timeline

Phase	Work	Duration
1	Literature and simulator setup	2–3 weeks
2	UAV dynamics and dataset generation	2–3 weeks
3	Physics calibration and baseline	2 weeks
4	Pure neural world-model baseline	2–3 weeks
5	Physics + residual model	3 weeks
6	Latent memory and open-loop imagination	3 weeks
7	Primary OOD wind experiments	3 weeks
8	Ablations and reliability analysis	3 weeks
9	Optional uncertainty / vision extension	2–3 weeks
10	Final analysis, writing	3–4 weeks

The minimum viable thesis is complete after Phase 7/8.

XVI. Expected Contributions

The expected contributions of the thesis are:

1) **Physics-prior latent world model:** A latent UAV dynamics architecture in which known rigid-body dynamics directly participate in the transition and a learned residual models unmodeled effects.
2) **Controlled long-horizon evaluation:** A systematic evaluation of recursive prediction error as a function of imagination horizon.
3) **OOD wind benchmark:** A controlled simulation protocol for evaluating learned UAV dynamics under wind conditions outside the training distribution.
4) **Reliable imagination horizon:** A physically interpretable metric describing how far into the future the model can maintain an acceptable level of prediction error.

5) **Analysis of when physics helps:** An empirical study of the interaction between physical-model accuracy, training-data quantity, distribution shift, and long-horizon prediction reliability.

The contribution is therefore primarily empirical and methodological: the thesis seeks to characterize the value and limitations of physical priors in latent UAV world models rather than claim that the individual components are themselves new.

XVII. Final Thesis Statement

The proposed thesis investigates whether known physical structure can improve the reliability of learned latent world models under distribution shift.

The central model is

$$\boxed{\dot{\hat{s}}_t = f_{\text{phys}}(\hat{s}_t, u_t) + d_\theta(h_t, \hat{s}_t, u_t).} \tag{78}$$

The resulting physical prediction is integrated and fed back into the latent transition:

$$\boxed{\hat{h}_{t+1} = F_\theta\left(h_t, \phi(\tilde{s}_{t+1}), u_t\right).} \tag{79}$$

The central evaluation quantity is the position prediction error

$$\boxed{E_p(H) = \frac{1}{H}\sum_{k=1}^{H} \left\|\hat{\mathbf{p}}_{t+k} - \mathbf{p}_{t+k}\right\|_2.} \tag{80}$$

The corresponding reliable imagination horizon is

$$\boxed{H_\epsilon = \max\left\{H : E_p(H) \le \epsilon\right\}.} \tag{81}$$

The central research question is therefore:

> *How far can a learned UAV world model reliably imagine under wind disturbances that were not observed during training, and can known quadrotor physics extend that reliable horizon?*

The thesis does not assume that the physics prior will always help. Instead, it seeks to determine whether, when, and why incorporating known physical structure changes the reliability of long-horizon imagined UAV trajectories.

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